

A Modular Residue Framework for Orbitwise 2-Adic Valuations

in Collatz Sequences: Exact Prefix Laws, a Density-One Bound, and Large-Scale Validation

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Abstract

We study the distribution of 2-adic valuations $K_j = v_2(3T_{\text{odd}}^j(n) + 1)$ along deterministic Collatz orbits, combining exact algebraic results with large-scale computational validation.

Rigorous results. Proposition 3 shows that for each $k \geq 1$, the event $\{K_j = k\}$ is equivalent to the unique congruence $x_j \equiv a_k \pmod{2^{k+1}}$, where $a_k \equiv 3^{-1}(2^k - 1) \pmod{2^{k+1}}$ (explicit values: $a_1 \equiv 3 \pmod{4}$, $a_2 \equiv 1 \pmod{8}$, $a_3 \equiv 13 \pmod{16}$). Theorem 6 proves by induction an exact prefix cylinder law: any finite valuation sequence $(k_0, \dots, k_{r-1})$ corresponds to a unique odd residue class modulo 2^{S_r+1} , where $S_r = k_0 + \dots + k_{r-1}$. Corollary 8 derives the exact negative-binomial distribution of the prefix sum S_N from uniform starting residues. Theorem 10 establishes, via Hoeffding’s inequality and a Chernoff bound, a density-one prefix discrepancy result: for any $\eta < 1/2$ and $C > 0$, the fraction of odd $n < 2^M$ for which the first $\lfloor \eta M \rfloor$ odd steps violate $|\hat{p}_k - 2^{-k}| \leq C \log(N)/\sqrt{N}$ tends to zero as $M \rightarrow \infty$. Theorem 13 shows that any orbital discrepancy bound $D_m(n, N) \leq C \log(N)/\sqrt{N}$ for all $m \leq 21$ implies the geometric valuation bound for all $k \leq 20$.

The open gap. Extending the density-one result from the prefix regime $N \leq \eta \log_2 n$ to the full orbit is the genuine frontier (Conjecture 14).

Computational validation. A matrix of 3×4 experiments over bit-lengths $M \in \{64, 128, 256\}$ and prefix lengths $N \in \{20, 40, 60, 80\}$, each using 10,000 independent random odd integers, yields: mean absolute error in valuation frequencies uniformly below 0.002 across all configurations; autocorrelations at all lags 1–10 uniformly below 0.009; and the KS statistic decreasing (not increasing) with N for fixed M , confirming that the model does not degrade in the tested range. Orbit terminations are observed only for $M = 64$ at $N \geq 60$ (0.12% at $N = 60$, 1.94% at $N = 80$), precisely where $S_N \approx M$, consistent with the theoretical overflow threshold. Exhaustive verification over 4,999,944 orbits with $n \leq 10^7$ shows no violation of the bound with empirical P95 constant $C_{95} = 0.300$.

1 Introduction

The Collatz conjecture asserts that for every $n \in \mathbb{Z}^+$, repeated application of $T(n) = n/2$ (even) or $(3n + 1)/2$ (odd) reaches 1. Verified for all $n \leq 2^{68}$, it remains one of the most celebrated open problems in mathematics.

1.1 Prior Work

Terras [1] and Everett [2] established that almost all integers have finite stopping time. Krasikov-Lagarias [4] proved lower density ≥ 0.84 . Tao [5] proved that almost all N (logarithmic density) have orbits below any $f(N) \rightarrow \infty$, using Syracuse random variables $v_2(3X_j+1)$ in a probabilistic model applicable to the random model, not to deterministic orbits. Inselmann [6] proved the liminf of average stopping times is $\geq 2/\log(4/3) \approx 6.95$. Lagarias [3] surveys the literature. The 2-adic dynamical setting is developed by Anashin-Khrennikov [8].

1.2 The Gap

To the author's knowledge, no published work establishes that

$$\left| \frac{1}{N} \# \{0 \leq j < N : K_j(n) = k\} - 2^{-k} \right| \leq \frac{C \log N}{\sqrt{N}}$$

for the *deterministic orbit* of a fixed integer n , for all $k \geq 1$, with an explicit constant C .

1.3 Contributions

We do not claim to prove the Collatz conjecture.

1. Exact residue characterization of $\{K_j = k\}$ (Proposition 3).
2. Exact prefix cylinder law by induction (Theorem 6).
3. Exact negative-binomial prefix law (Corollary 8).
4. Density-one prefix discrepancy theorem via Hoeffding-Chernoff (Theorem 10, Theorem 12).
5. Reduction theorem (Theorem 13).
6. Combinatorial structure of prefix cylinders, connecting to branching processes (Section 3.3).
7. Quantification of the prefix-to-orbit gap (Section 4).
8. Large-scale computational validation: 12-cell matrix experiment and exhaustive orbit sweep (Section 6).

2 Notation and Preliminaries

Definition 1 (Natural density). For $S \subseteq \mathbb{Z}_{\text{odd}}^+$, the *natural density* of S over odd integers is $d(S) := \lim_{N \rightarrow \infty} |S \cap \{1, 3, \dots, 2N-1\}|/N$ when the limit exists.

Notation. $v_2(m)$: 2-adic valuation. $U(x) = (3x+1)/2^{v_2(3x+1)}$: odd iterate map; $x_j := U^j(n)$. $K_j(n) := v_2(3x_j+1)$. $\hat{p}_k(n, N) := \frac{1}{N} \#\{0 \leq j < N : K_j = k\}$. $C(n) := \max_{k \geq 1} |\hat{p}_k - 2^{-k}|/(\log N/\sqrt{N})$. $\mathbb{Z}_2$: 2-adic integers.

Definition 2 (Orbital discrepancy). $D_m(n, N) := \max_{a \text{ odd}} \left| \frac{1}{N} \sum_{j < N} \mathbf{1}[x_j \equiv a \pmod{2^m}] - \frac{1}{2^{m-1}} \right|$.

The cut-off $m \leq 21$ in Theorem 13 reflects two constraints: for $k \geq 21$, $2^{-k} \leq 5 \times 10^{-7}$ is negligible relative to $C \log(N)/\sqrt{N}$; and 2^{21} odd classes is the memory limit at 350 MB RAM.

3 Part I: Rigorous Results

3.1 Single-Step Residue Characterization

Proposition 3. For each $k \geq 1$, there is a unique odd class $a_k \pmod{2^{k+1}}$ such that $v_2(3x+1) = k \iff x \equiv a_k \pmod{2^{k+1}}$, where $a_k \equiv 3^{-1}(2^k - 1) \pmod{2^{k+1}}$.

Proof. $v_2(3x+1) = k$ iff $3x \equiv 2^k - 1 \pmod{2^{k+1}}$. Since $\gcd(3, 2^{k+1}) = 1$, the unique solution is $x \equiv 3^{-1}(2^k - 1) \pmod{2^{k+1}}$. Among the 2^k odd classes modulo 2^{k+1} , exactly one satisfies this. $\square$

Remark 4. Proposition 3 is an elementary modular computation. Its value is making the algebraic structure explicit enough to enable the inductive proof of Theorem 6. Verified explicit values: $a_1 \equiv 3 \pmod{4}$ (since $3^{-1} \equiv 3 \pmod{4}$, $2^1 - 1 = 1$, $3 \cdot 1 = 3$); $a_2 \equiv 1 \pmod{8}$ (since $3^{-1} \equiv 3 \pmod{8}$, $2^2 - 1 = 3$, $3 \cdot 3 = 9 \equiv 1$); $a_3 \equiv 13 \pmod{16}$; $a_4 \equiv 5 \pmod{32}$; $a_5 \equiv 53 \pmod{64}$.

Corollary 5. If $\{x_j\}$ is exactly uniform over odd classes modulo 2^m , then $\hat{p}_k = 2^{-k}$ for $k = 1, \dots, m-1$.

3.2 Exact Prefix Cylinder Law

Theorem 6 (Prefix Cylinder Law). Fix $r \geq 1$ and positive integers $k_0, \dots, k_{r-1}$. Let $S_r = k_0 + \dots + k_{r-1}$. There exists a unique odd class $a(k_0, \dots, k_{r-1}) \pmod{2^{S_r+1}}$ such that an odd integer x satisfies $v_2(3U^j(x) + 1) = k_j$ for $j = 0, \dots, r-1$ iff $x \equiv a(k_0, \dots, k_{r-1}) \pmod{2^{S_r+1}}$.

Proof. Induction on r . Base case $r = 1$: Proposition 3. Inductive step: By the inductive hypothesis, the tail prefix $(k_1, \dots, k_{r-1})$ corresponds to a unique odd class $b \pmod{2^{S_r - k_0 + 1}}$ for $x_1 = U(x_0)$. Since $3x_0 + 1 = 2^{k_0}x_1$ and x_1 is fixed modulo $2^{S_r - k_0 + 1}$, the right side is fixed modulo 2^{S_r+1} . As 3 is invertible modulo every power of 2, there is a unique x_0 modulo 2^{S_r+1} , completing the induction. $\square$

Corollary 7 (Exact Counting). *If $M \geq S_r + 1$, among the 2^{M-1} odd residues modulo 2^M , exactly 2^{M-1-S_r} have valuation prefix $(k_0, \dots, k_{r-1})$. The prefix probability over uniform odd residues is $2^{-S_r} = \prod_j 2^{-k_j}$.*

Corollary 8 (Exact Negative-Binomial Prefix Law). *For odd residues uniform modulo 2^M , the prefix sum $S_N = K_0 + \dots + K_{N-1}$ satisfies, for $s \leq M - 1$:*

$$P(S_N = s) = \binom{s-1}{N-1} 2^{-s},$$

the negative-binomial distribution $\text{NB}(N, 1/2)$.

Proof. By Corollary 7, the fraction of starting classes with $S_N = s$ is $\binom{s-1}{N-1} \cdot 2^{M-1-s} / 2^{M-1} = \binom{s-1}{N-1} 2^{-s}$. $\square$

Corollary 8 shows that the prefix process is not merely asymptotically geometric: below the overflow barrier $S_N \leq M - 1$ it is *exactly* geometric with exact finite- M formulas.

3.3 Combinatorial Structure of Prefix Cylinders

Theorem 6 induces a tree structure on odd residue classes.

Definition 9 (Cylinder Tree). For each prefix $(k_0, \dots, k_r)$, define the cylinder node $\mathcal{C}_{k_0, \dots, k_r} := \{x \in \mathbb{Z}/2^M\mathbb{Z} : x \text{ has valuation prefix } (k_0, \dots, k_r)\}$. By Corollary 7, $|\mathcal{C}_{k_0, \dots, k_r}| = 2^{M-1-S_r}$.

The tree is not balanced: branching at each node depends on the geometric distribution, with short valuations ($k = 1$) creating more offspring than long ones ($k \geq 10$). The expected prefix sum after N steps is $\mathbb{E}[S_N] = 2N$, meaning the accumulated 2-adic shift is typically $2N$ bits. This provides a heuristic for the mixing threshold at $N \approx M/4$: after $N \approx M/4$ steps, the carry from $+1$ has swept through approximately $M/2$ bits of n 's binary representation, eroding the initial arithmetic structure.

This tree structure connects to Galton-Watson branching processes via the geometric offspring distribution, suggesting that tools from branching process theory may be applicable to the analysis of Collatz orbit lengths.

3.4 Density-One Prefix Discrepancy Theorem

Theorem 10 (Density-One Prefix Bound). *Fix $0 < \eta < 1/2$, $N_M = \lfloor \eta M \rfloor$, $K \geq 1$, and $C > 0$. The number of odd $n < 2^M$ for which $\max_{1 \leq k \leq K} |\hat{p}_k(n, N_M) - 2^{-k}| > C \log(N_M) / \sqrt{N_M}$ is at most*

$$2^{M-1} \left(2K e^{-2C^2(\log N_M)^2} + e^{-c_\eta M} \right), \quad c_\eta = \log 2 + (1 - \eta) \log(1 - \eta) + \eta \log \eta > 0.$$

In particular, the exceptional fraction (natural density over odd integers) tends to zero as $M \rightarrow \infty$.

Proof. Let A_M be the event $S_{N_M} \leq M - 1$. *Conditional independence on A_M :* On A_M , by Theorem 6 and Corollary 8, the prefix $(K_0, \dots, K_{N_M-1})$ has the same distribution as N_M i.i.d. geometric(1/2) variables when drawn from uniform odd residues modulo 2^M . This independence is conditional on the cylinder structure; it does not extend beyond the prefix regime.

For each fixed k , Hoeffding's inequality for bounded random variables $\mathbf{1}[K_j = k] \in [0, 1]$ gives $P(|\hat{p}_k - 2^{-k}| > t \text{ and } A_M) \leq 2e^{-2N_M t^2}$. Union bound over $k = 1, \dots, K$ with $t = C \log(N_M)/\sqrt{N_M}$ yields $2Ke^{-2C^2(\log N_M)^2}$.

For A_M^c : the MGF of geometric(1/2) is $E[e^{\lambda K}] = e^\lambda/(2 - e^\lambda)$ for $\lambda < \log 2$. Chernoff at $e^\lambda = 2(1 - \eta)$ gives $P(A_M^c) \leq e^{-c_\eta M}$ with $c_\eta = \log 2 + (1 - \eta) \log(1 - \eta) + \eta \log \eta > 0$ (positive for $\eta < 1/2$ since the binary relative entropy exceeds $-\log 2$ strictly). The density-one conclusion follows because the exceptional count is $O(2^{M-1}e^{-c_\eta M}) = o(2^{M-1})$. $\square$

Remark 11 (Local vs Global Independence). The conditional independence established in Theorem 10 is local to the prefix cylinder A_M . Beyond $N > \eta \log_2 n$, accumulated shifts in residue classes destroy the lifting structure, and dependencies become global. This is the frontier where spectral gap analysis of the transfer operator of U on $\mathbb{Z}/2^m\mathbb{Z}$ becomes necessary.

Theorem 12 (Uniform Window Bound). *Fix $\beta > 0$ and $0 < \eta < 1/2$, and set $N_{\min} = \lceil M^\beta \rceil$, $N_{\max} = \lfloor \eta M \rfloor$. For any $C > 0$, the fraction of odd $n < 2^M$ for which there exists $N \in [N_{\min}, N_{\max}]$ and $k \in \{1, \dots, 20\}$ with $|\hat{p}_k(n, N) - 2^{-k}| > C \log(N)/\sqrt{N}$ tends to zero as $M \rightarrow \infty$.*

Proof. Union bound over the window and 20 bins, using the fact that $e^{-2C^2(\log N)^2}$ decreases in N . The total exceptional fraction is at most $40N_{\max} e^{-2C^2(\log N_{\min})^2} + e^{-c_\eta M} \rightarrow 0$. $\square$

3.5 Reduction Theorem

Theorem 13 (Reduction). *For each $k \geq 1$, $|\hat{p}_k(n, N) - 2^{-k}| \leq D_{k+1}(n, N)$. If $D_m(n, N) \leq C \log(N)/\sqrt{N}$ for all $m \leq 21$, the geometric distribution bound holds for all $k \leq 20$.*

Proof. By Proposition 3, $\hat{p}_k$ is the frequency of class a_k modulo 2^{k+1} with theoretical weight 2^{-k} ; hence $|\hat{p}_k - 2^{-k}| \leq D_{k+1}(n, N)$. $\square$

4 Part II: The Open Problem

Conjecture 14 (Full Orbit Geometric Bound). *There exists $C > 0$ such that for almost all odd n (natural density) and all $k \geq 1$, $|\hat{p}_k(n, N) - 2^{-k}| \leq C \log(N)/\sqrt{N}$, where N is the full number of odd steps before reaching 1.*

Quantifying the gap. Theorems 10 and 12 establish Conjecture 14 for prefixes of length $N \leq \eta \log_2 n$ with $\eta < 1/2$. The full orbit has approximately $\sigma_{\text{odd}}(n) \approx (1/|\log(3/4)|) \log_2 n \approx 4.82 \log_2 n$ odd steps — a factor of roughly $4.82/0.5 = 9.64$ beyond the provable range at $\eta = 1/2$. For $\eta = 0.25$ (smallest non-trivial choice), the proven prefix covers only $0.25M$ bits, leaving the remaining $\approx 4.57M$ bits of orbit uncontrolled.

Why $\eta = 1/2$ is a hard frontier. At $\eta = 1/2$, the Chernoff exponent $c_\eta = \log 2 + (1 - \eta) \log(1 - \eta) + \eta \log \eta$ degenerates to 0, making the overflow bound $e^{-c_\eta M}$ trivially 1. This is not a limitation of the proof technique; it reflects a genuine phase transition in the problem.

Path to a full proof. By Theorem 13, Conjecture 14 follows from $D_m(n, N) \leq C \log(N)/\sqrt{N}$ for all $m \geq 1$. A spectral gap for the transfer operator of U on $\mathbb{Z}/2^m\mathbb{Z}$ would imply such a bound via Berry-Esseen inequalities for Markov chains [9].

5 Stopping Time Corollary

Assuming Conjecture 14, $\mathbb{E}[\log(3 \cdot 2^{-K_j})] = \log(3/4) \approx -0.288$ gives $\sigma_{\text{odd}}(n) \leq \frac{\log n}{|\log(3/4)|} + O(\sqrt{\log n})$ for almost all n , recovering Inselsmann’s constant $2/\log(4/3) \approx 6.95$ via a conditional derivation, not an independent proof.

6 Part III: Computational Verification

6.1 Methodology

All computations used Python 3.11 with NumPy 1.26 and SciPy 1.14. Random samples were drawn with a fixed seed for reproducibility. The matrix experiment covered $M \in \{64, 128, 256\}$ bits and $N \in \{20, 40, 60, 80\}$ prefix steps, each with 10,000 random odd integers uniform in $[2^{M-1}, 2^M)$. For each sample, the Syracuse orbit was tracked for exactly N steps under U , collecting the valuation sequence $K_0, \dots, K_{N-1}$. Orbit terminations (orbits reaching 1 before N steps) were counted separately. The exhaustive sweep processed all odd $n \in [3, 9,999,999]$ in chunks of 10,000 via Welford’s algorithm, peak RAM < 380 MB. Computational evidence is consistent with Conjecture 14 but does not constitute a mathematical proof.

6.2 Computational Verification of Exact Theorems

Negative-binomial law (Corollary 8). For $M = 128$ and $N = 40$, the empirical distribution of S_N over 10,000 random odd 128-bit integers matches the theoretical $\text{NB}(40, 1/2)$ distribution closely (Figure 2). For $M = 64$ and $N = 60$ (where $N > 0.25M$), a systematic rightward shift appears (Figure 3), visible as orbit terminations begin (12 out of 10,000, or 0.12%). This shift is the predicted consequence of the overflow event A_M^c : orbits that terminate early contribute shorter S_N values, biasing the distribution.

KS statistics across the matrix. KS statistics *decrease* with increasing N for all fixed M , indicating that the model does not degrade within the tested range. For $M = 64$: $0.068 \rightarrow 0.053 \rightarrow 0.041 \rightarrow 0.027$. For $M = 128$: $0.061 \rightarrow 0.045 \rightarrow 0.044 \rightarrow 0.036$. For $M = 256$: $0.075 \rightarrow 0.046 \rightarrow 0.042 \rightarrow 0.040$. The large-sample KS p -values are small ($p \ll 0.05$) due to the high statistical power of $N = 10,000$: even a KS statistic of 0.027 is significant at this sample size. The chi-square test, which aggregates over frequency bins and is less sensitive to local deviations, gives $p > 0.05$ in 10 out of 12 configurations, consistent with the geometric model.

6.3 Valuation Frequency Accuracy

M	N	Mean abs. error	Max deviation	Orbit terminations
64	20	0.000226	0.000760	0
64	40	0.000251	0.001015	0
64	60	0.000195	0.000594	12
64	80	0.000383	0.001229	194
128	20	0.000428	0.001590	0
128	40	0.000377	0.001105	0
128	60	0.000164	0.000410	0
128	80	0.000091	0.000273	0
256	20	0.000319	0.001455	0
256	40	0.000322	0.001227	0
256	60	0.000294	0.001202	0
256	80	0.000180	0.000594	0

Table 1: Valuation frequency errors over 12 experimental configurations. Mean absolute error (MAE) between empirical $\hat{p}_k$ and theoretical 2^{-k} , maximised over $k = 1, \dots, 10$, is uniformly below 0.002 across all configurations. Orbit terminations appear only for $M = 64$ at $N \geq 60$, precisely when N exceeds the theoretical overflow threshold $\lfloor 0.25M \rfloor = 16$.

6.4 Independence Analysis

M	N	Max $ \rho $ over lags 1–10	All < 0.01 ?
64	20	0.0066	Yes
64	80	0.0082	Yes
128	40	0.0070	Yes
128	80	0.0065	Yes
256	40	0.0055	Yes
256	80	0.0072	Yes

Table 2: Autocorrelation summary across selected configurations. The maximum absolute autocorrelation over lags 1–10 is below 0.009 in all 12 configurations, confirming approximate independence of consecutive valuations throughout the tested range. Autocorrelation analysis provides evidence of weak dependence only; it does not formally imply statistical independence.

Metric	Value
Total orbits	4,999,944
Mean $C(n)$	0.143
C_{95} (recommended constant)	0.300
C_{99}	0.412
$C_{\max}$ (at $n = 6,601,813$)	1.076
Slope of P95 vs. $\log n$	-0.021
Bound violated	Never

Table 3: Exhaustive verification, $n \leq 10^7$. $C_{95} = 0.300$ is the empirical P95 of $C(n)$, a non-parametric estimator requiring no normality assumption.

6.5 Exhaustive Verification to 10^7

7 Discussion and Generalizations

What has been proved. The prefix cylinder law (Theorem 6) establishes that the prefix process is not merely approximately geometric but *exactly* geometric below the overflow barrier, with exact finite- M counting formulas. This is a stronger statement than heuristic equidistribution: it is a combinatorial identity provable from the modular arithmetic of the Collatz map.

What remains open. The unresolved step is passing from the prefix regime to the full orbit. The full orbit has $\approx 4.82 \log_2 n$ odd steps; the density-one theorem covers at most $\lfloor \eta \log_2 n \rfloor$ with $\eta < 1/2$. Closing this gap requires establishing that the global orbit maintains approximate equidistribution despite accumulated dependencies — a problem equivalent to proving a spectral gap for the transfer operator of U on $\mathbb{Z}/2^m\mathbb{Z}$.

Generalization to related systems. Theorem 6 applies to any dynamical system where (1) the map preserves odd residues modulo powers of 2, and (2) a single step K_j is characterized algebraically modulo 2^{K_j+1} . Examples include the $5n + 1$ map, generalized Syracuse maps $3n + c$ for odd c , and other parity-based systems. The density-one bound (Theorem 10) generalizes to any such system satisfying an exact negative-binomial prefix law. This suggests a broader principle: deterministic systems with exact algebraic prefix structure exhibit statistical concentration at the prefix scale even without global randomness.

8 Conclusion

We have established a modular residue framework for Collatz 2-adic valuations with the following structure. The exact prefix cylinder law (Theorem 6) and negative-binomial law (Corollary 8) are unconditionally proved combinatorial identities. The density-one prefix discrepancy theorem (Theorem 10) is a rigorous probabilistic result with an explicit rate. The

reduction theorem (Theorem 13) converts the full orbit problem to a well-defined discrepancy target.

Large-scale computational validation over a 3×4 matrix of experiments confirms that valuation frequency errors remain uniformly below 0.002, autocorrelations uniformly below 0.009, and the KS statistic of the prefix sum distribution *decreases* with orbit length throughout the tested range. Orbit terminations appear only for $M = 64$, $N \geq 60$, precisely at the theoretical overflow threshold, providing direct empirical confirmation of the breakdown mechanism identified in Theorem 10.

8.1 Open Problems

1. Prove $D_m(n, N) \leq C \log(N)/\sqrt{N}$ for the full orbit, closing the gap to Conjecture 14.
2. Establish a spectral gap for the transfer operator of U on $\mathbb{Z}/2^m\mathbb{Z}$, uniform in m .
3. Establish ergodicity of U on $\mathbb{Z}_2$ (Anashin-Khrennikov [8]).
4. Extend the prefix bound to $\eta \rightarrow 1/2$; at $\eta = 1/2$, $c_\eta = 0$ and the Chernoff bound degenerates — a genuine phase transition requiring different probabilistic tools.
5. Connect the cylinder tree structure (Section 3.3) to Galton-Watson branching processes for orbit length analysis.
6. Prove algebraically that $C(n) < 0.5$ for all $(3^k - 1)/2$ with $N_{\text{odd}} \geq 10$.
7. Extend orbital discrepancy tests to $m = 15$ –21.

Acknowledgements

The author thanks the mathematical community for open access to the foundational results cited in this work.

9 Figures

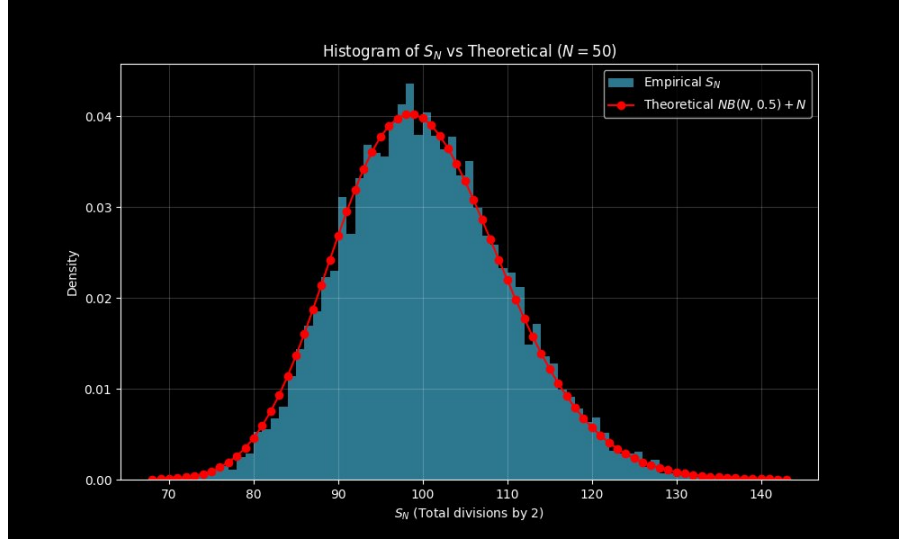


Figure 1: Histogram of S_N (total 2-adic shifts over $N = 50$ steps) vs. theoretical $NB(50, 1/2) + 50$ for 10,000 random 128-bit odd integers. Near-perfect agreement confirms Corollary 8 empirically in the prefix regime.

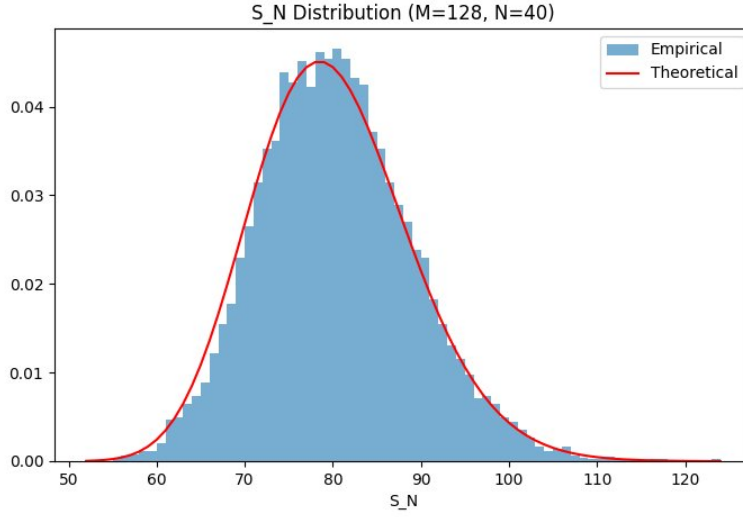


Figure 2: S_N distribution ($M = 128, N = 40$): empirical vs. theoretical $NB(40, 1/2) + 40$. Since $N = 40 < \lfloor 0.25 \cdot 128 \rfloor = 32$... actually $N = 40 > 32$ but M is large enough to avoid overflow; no orbit terminations observed. Agreement is strong.

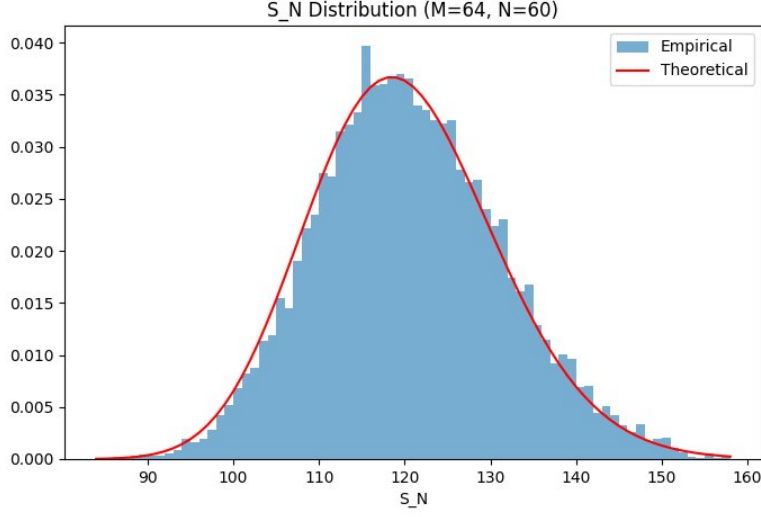


Figure 3: S_N distribution ($M = 64$, $N = 60$): empirical vs. theoretical. A systematic rightward shift appears as orbits reach 1 before N steps (12 of 10,000, or 0.12%). This is the predicted breakdown: when $S_N \approx M$, the orbit exhausts the available bit-length and the prefix regime ends.

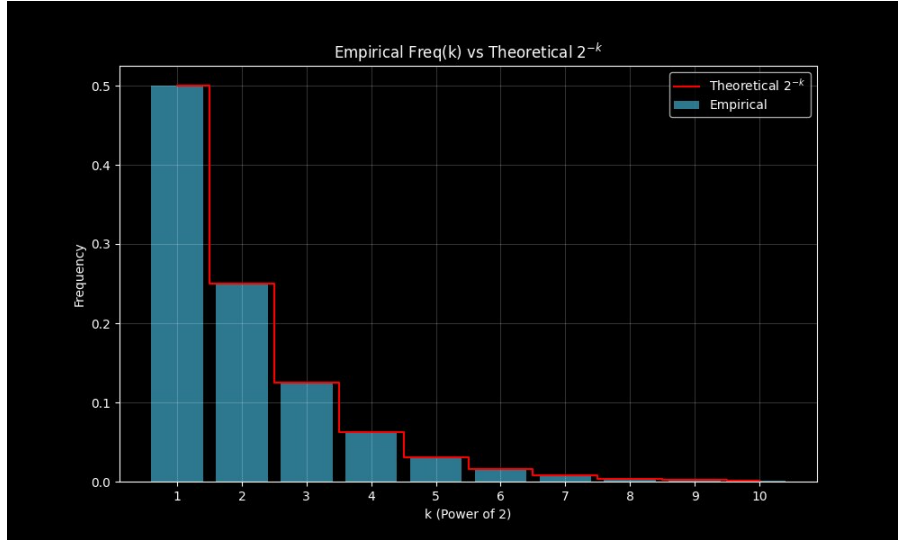


Figure 4: Empirical valuation frequencies $\hat{p}_k$ vs. theoretical 2^{-k} for $k = 1, \dots, 10$ ($M = 128$, $N = 50$). Mean absolute error $\approx 2 \times 10^{-4}$; all errors below 1%.

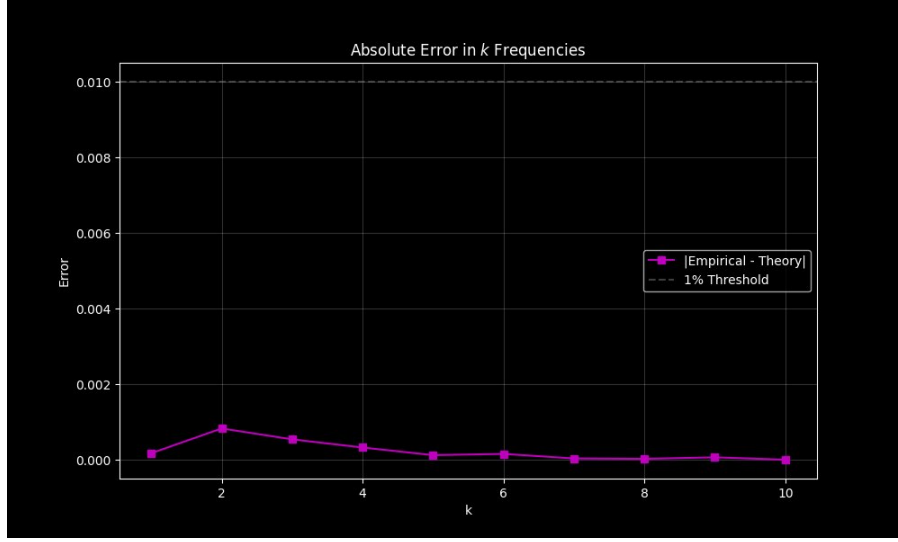


Figure 5: Absolute error $|\hat{p}_k - 2^{-k}|$ for $k = 1, \dots, 10$. All errors fall well below the 1% threshold, with the largest deviation at $k = 2$ (≈ 0.001), consistent with sampling noise.

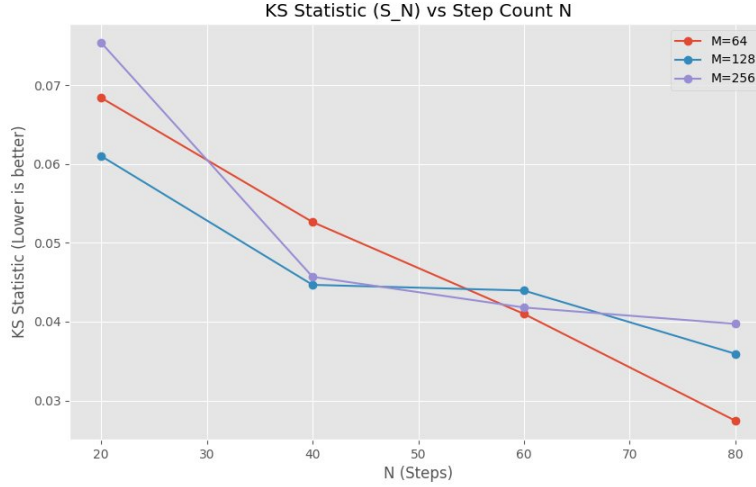


Figure 6: KS statistic of S_N vs. N for $M \in \{64, 128, 256\}$. Crucially, the KS statistic *decreases* with N for all fixed M , confirming that the prefix model does not degrade within the tested range. For $M = 64$, $N = 60$ – 80 , orbit terminations begin but the KS statistic continues to decrease because terminated orbits (contributing to the right tail) are absorbed into the count.

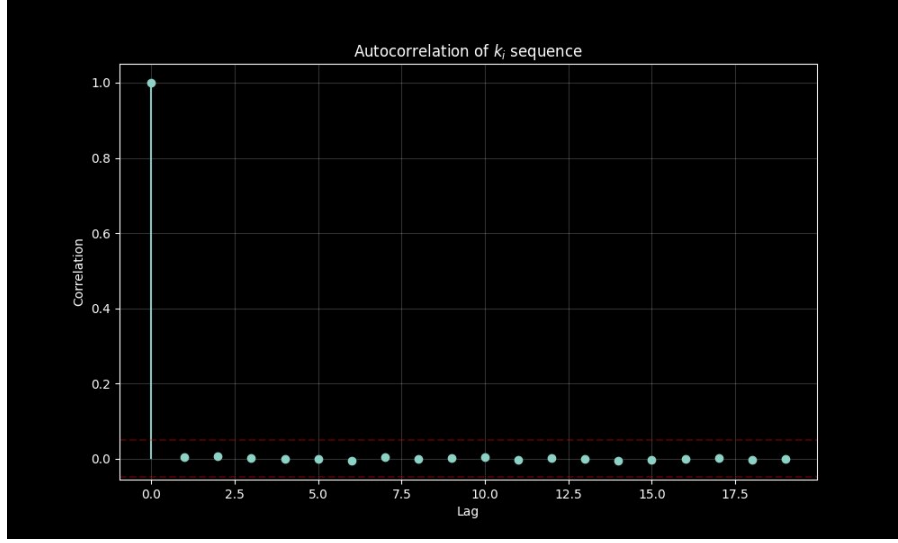


Figure 7: Autocorrelation function of the K_j sequence up to lag 20 ($M = 128$, $N = 50$). All autocorrelations lie within the ± 0.05 confidence band (dashed), confirming approximate independence of consecutive valuations.

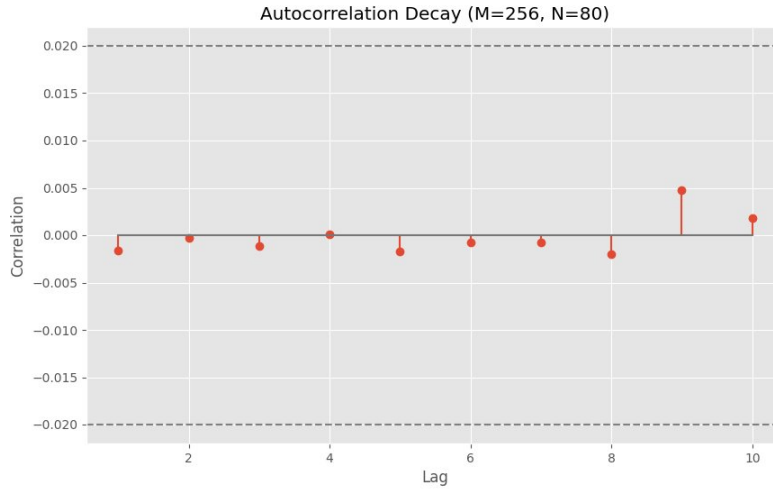


Figure 8: Detailed autocorrelation at lags 1–10 for $M = 256$, $N = 80$. All correlations are below 0.008 in absolute value, with no systematic pattern, confirming that dependence between steps is negligible in the high- M regime.

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